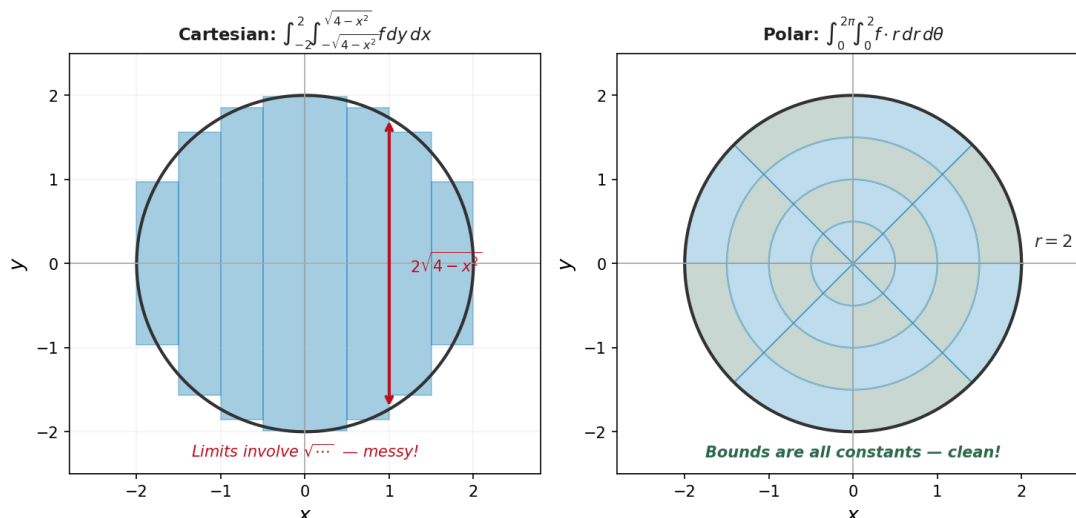


Why Polar?

Not every region fits nicely into Cartesian coordinates. Consider integrating over a disk like $x^2 + y^2 \leq 4$.



In Cartesian, the bounds are:

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} f(x, y) dy dx$$

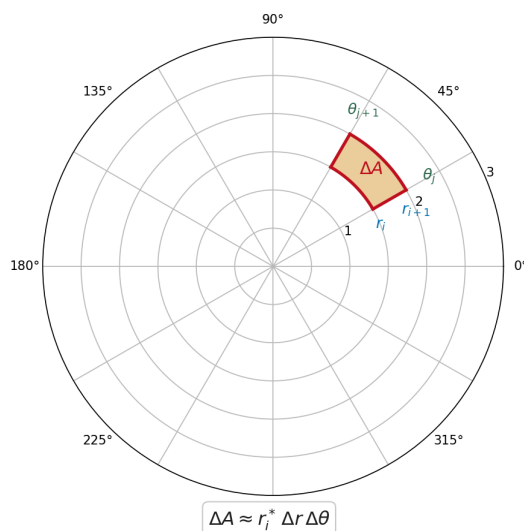
That square root makes everything painful — and the region isn't x -simple or y -simple in a clean way. But in **polar coordinates**, a disk of radius 2 is just $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$. The bounds are **constants**!

Regions described by circles, sectors, and cardioids are natural in polar. When the region has circular symmetry, polar coordinates will simplify the integral dramatically.

Polar Rectangles and the Area Element

In Cartesian coordinates, we chop the plane into small rectangles of area $\Delta A = \Delta x \Delta y$.

In polar coordinates, we chop the plane into **polar rectangles**: regions bounded by two radii and two arcs.



A polar rectangle with r from r_i to $r_i + \Delta r$ and θ from θ_j to $\theta_j + \Delta \theta$ has area:

$$\Delta A_i \approx r_i \Delta r \Delta \theta$$

The key insight: polar rectangles are **wider** at larger r . The factor of r accounts for this stretching. A small $\Delta r \Delta \theta$ box near $r = 10$ covers more area than the same box near $r = 1$.

In polar coordinates, $dA = r dr d\theta$ — **not** just $dr d\theta$. The extra r is essential.

The Polar Double Integral

Double Integral in Polar Coordinates

If f is continuous on a polar region R described by $\alpha \leq \theta \leq \beta$ and $h_1(\theta) \leq r \leq h_2(\theta)$, then:

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$$

Three things change when converting to polar:

1. Replace x with $r \cos \theta$ and y with $r \sin \theta$
2. Replace dA with $r dr d\theta$ (don't forget the r !)
3. Replace the Cartesian bounds with polar bounds

The most common mistake is forgetting the extra r in $dA = r dr d\theta$. Write it down every time.

Quick Check: Convert to Polar

Set up (but **do not evaluate**) the polar integral for:

$$\iint_R (x^2 + y^2) dA$$

where R is the unit disk $x^2 + y^2 \leq 1$.

What replaces $x^2 + y^2$? What replaces dA ? What are the bounds?

Use $x^2 + y^2 = r^2$ and $dA = r dr d\theta$. The unit disk is $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$.

Example 1: Area of $r = 4 \cos \theta$ in QI

Example 1

Find the area in Quadrant I enclosed by the graph of $r = 4 \cos \theta$.

Area = $\iint_R 1 \, dA$ in polar. What does $r = 4 \cos \theta$ look like? Where does it start and end? What are the bounds on r and θ ?

Example 2: Volume of a Hemisphere

Example 2

Evaluate $\iint_R \sqrt{4 - x^2 - y^2} \, dA$ where $R = \{(x, y) \mid x^2 + y^2 \leq 4, x \geq 0\}$.

Convert to polar: $\sqrt{4 - x^2 - y^2} = \sqrt{4 - r^2}$. What geometric shape is R ? What are the polar bounds on r and θ ?

Quick Check: Sketch a Polar Region

Sketch the region described by $1 \leq r \leq 2$ and $0 \leq \theta \leq \pi/4$. What shape is it?

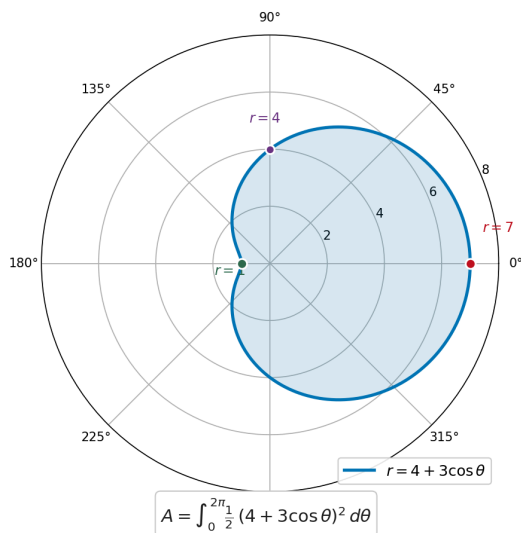
Mark the radii $r = 1$ and $r = 2$ and the angles $\theta = 0$ and $\theta = \pi/4$. Shade the region between them.

Example 3: Area of the Limaçon $r = 4 + 3 \cos \theta$

Example 3

Find the area enclosed by $r = 4 + 3 \cos \theta$.

Since $4 > 3$, this is a limaçon **without an inner loop** — it's a slightly flattened, egg-shaped curve. The curve is traced once as θ goes from 0 to 2π .



Area = $\int_0^{2\pi} \int_0^{4+3 \cos \theta} r dr d\theta$. Evaluate the inner integral first, then use the identity $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$.

Example 4: Volume Inside a Sphere and Outside a Cylinder

Example 4

Find the volume of the solid inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.

Use symmetry (double the top half). Convert to polar. What are the bounds on r ? On θ ? Don't forget $dA = r \, dr \, d\theta$.

When to Use Polar Coordinates

How do you know when to switch to polar? Here are the telltale signs:

Clue	Example
Region is a disk, annulus, or sector	$x^2 + y^2 \leq 9$
Boundary involves $x^2 + y^2$	$1 \leq x^2 + y^2 \leq 4$
Integrand contains $x^2 + y^2$	$e^{-(x^2+y^2)}$
Boundary is a polar curve	$r = 1 + \cos \theta$
Square-root bounds like $\sqrt{a^2 - x^2}$	Semicircle or quarter-circle region

Conversion checklist:

- 1. Replace $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$
- 2. Replace $dA = dy \, dx$ with $r \, dr \, d\theta$ (don't forget the r !)
- 3. Determine polar bounds for r and θ
- 4. Evaluate (inner r first, then outer θ)

Not every integral is easier in polar. If the region is a rectangle or the integrand involves x and y separately (not in $x^2 + y^2$ combinations), Cartesian may be simpler.

Homework

Section 15.3: 1-7, 9, 11, 13, 17, 19, 21, 23, 25, 27, 29, 31, 50a